

2024 春电路电子学参考答案

一、计算题（24 分）

1. （6 分）

$$\text{节点①: } (\frac{1}{2\Omega} + \frac{1}{2\Omega} + \frac{1}{1\Omega})U_{n1} - \frac{1}{1\Omega}U_{n2} - \frac{1}{2\Omega}U_{n3} = \frac{4V}{2\Omega} + 3A$$

$$\text{节点②: } -\frac{1}{1\Omega}U_{n1} + (\frac{1}{4\Omega} + \frac{1}{1\Omega})U_{n2} - \frac{1}{4\Omega}U_{n3} = -2I_1$$

$$\text{节点③: } U_{n3} = 10V$$

$$I_1 = \frac{U_{n1}}{2\Omega}$$

$$\text{解得: } U_{n1} = 6V, U_{n2} = 2V, U_{n3} = 10V$$

$$I_1 = \frac{U_{n1}}{2\Omega} = 3A$$

2. （6 分）

(1)求开路电压 U_{OC}

由 KCL, $I - 2I = 0$, 得 $I = 0$

所以开路电压 $U_{OC} = -20I + 8 = 8V$

(2)求等效电阻

求短路电流, $10 \times 2I = 8$, 解得 $I = 0.4A$; 短路电流 $I_{SC} = 2I - I = 0.4A$

$$\text{等效电阻 } R_i = \frac{U_{OC}}{I_{SC}} = \frac{8}{0.4} = 20\Omega$$

$$R_L = 20\Omega, P_{\max} = \frac{U_{OC}^2}{4R_i} = \frac{8^2}{4 \times 20} = 0.8W$$

3. （6 分）

$$\text{初始值 } i_L(0_+) = i_L(0_-) = \frac{4}{4+1} \times 5mA = 4mA$$

$$\text{稳态值 } i_L(\infty) = \frac{4}{4+4} \times 5 = 2.5mA$$

$$\text{等效电阻 } R_i = 4 + 1 + 3 = 8k\Omega$$

$$\text{时间常数 } \tau = \frac{L}{R_i} = \frac{0.8}{8 \times 10^3} = 10^{-4}s$$

$$\text{由三要素公式得: } i_L(t) = i_L(\infty) + [i_L(0_+) - i_L(\infty)]e^{-t/\tau} = [2.5 + 1.5e^{-10^4 t}]mA \quad (t > 0)$$

4. （6 分）

因为电路处于谐振状态, 故电感与电容串联电路相当于短路, 因此有

$$\frac{R_1 R_2}{R_1 + R_2} = \frac{U_1}{I_s} = 50\Omega$$

代入 $R_1 = 100\Omega$ ，解得 $R_2 = 100\Omega$

又因为电路处于谐振状态，所以 $X_L = |X_C| = 100\Omega$

$$\text{故有 } U_L = I_2 X_L = \frac{R_1 I_s}{R_1 + R_2} \times X_L = 50V$$

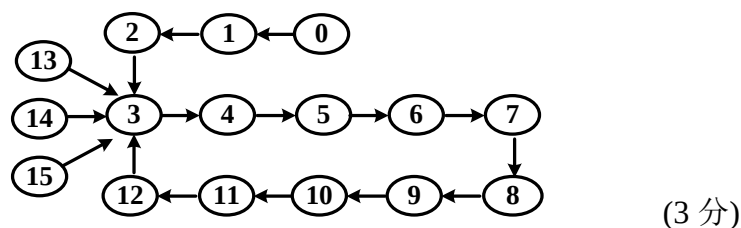
二、设计题（14分）

$$1. F = \bar{A}\bar{B}C + AB\bar{C} + ABC = \sum m(5, 6, 7) \quad (2 \text{ 分})$$

$$D_0 = D_1 = D_2 = D_3 = D_4 = 0 \quad (2 \text{ 分})$$

$$D_5 = D_6 = D_7 = 1 \quad (2 \text{ 分})$$

2. 十进制 (2分)



余三码 (1分)

$$B_4 B_3 B_2 B_1 = 1101 \quad (2 \text{ 分})$$

三、分析题（32分）

1. 交直流电流串联负反馈 (4分)

$$2. F = A + BC + BD \quad (4 \text{ 分})$$

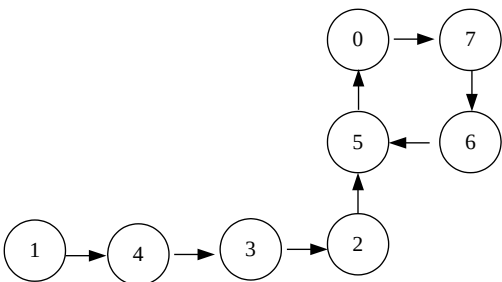
$$3. J_0 = K_0 = 1 \quad J_1 = K_1 = \bar{Q}_0 \quad J_2 = \bar{Q}_0 \bar{Q}_1 \quad K_2 = \bar{Q}_1 \quad (3 \text{ 分})$$

$$Q_0^{n+1} = \bar{Q}_0^n \quad Q_1^{n+1} = Q_0^n \odot Q_1^n \quad Q_2^{n+1} = \bar{Q}_0^n \bar{Q}_1^n \cdot \bar{Q}_2^n + Q_1^n Q_2^n \quad (3 \text{ 分})$$

状态转换表

态序	Q_2	Q_1	Q_0
0	0	0	0
1	1	1	1
2	1	1	0
3	1	0	1
4	0	0	0
0	0	0	1
1	1	0	0
2	0	1	1
3	0	1	0
4	1	0	1

(1 分)

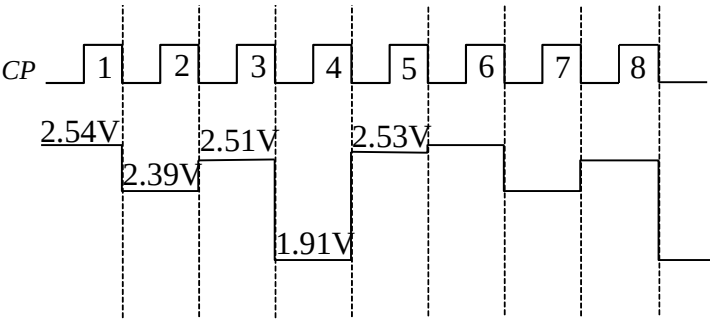


(1 分)

四进制 (1 分)

能自启动 (1 分)

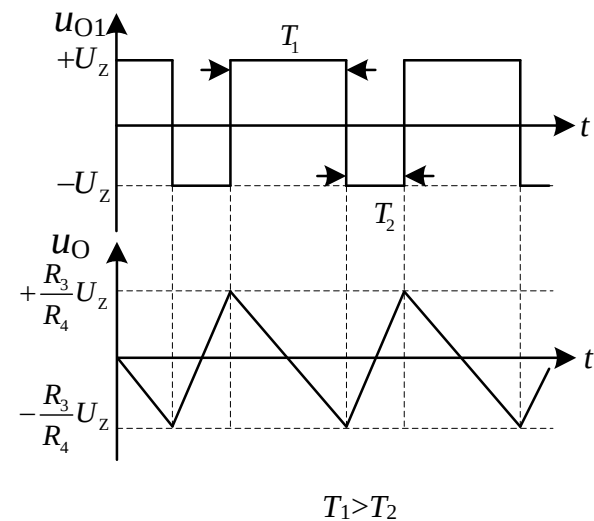
4. (6 分)



5.

$$u_{om} = \frac{R_3}{R_4} U_z \quad (2 \text{ 分})$$

$$T = \frac{4R_3R_2C}{R_4} \quad (2 \text{ 分})$$



(4 分)